1 Parameterized Quarto Reports

1.1 Learning objectives

After completing this chapter, you will be able to:

- Define parameters in a Quarto document's YAML header
- Access parameters in R code chunks to drive data retrieval and analysis
- Render parameterized reports from the command line and from scripts
- Build a template for journal-level bibliometric profiles
- Generate multiple reports in batch using purrr or shell loops

1.2 Setup

```
library(tidyverse)
library(openalexR)
library(glue)
library(gt)

set.seed(20260509)

source(here::here("R", "api_helpers.R"))
source(here::here("R", "utils.R"))
source(here::here("R", "sci_palette.R"))
```

1.3 Conceptual background

Many bibliometric tasks are repetitive: generating the same analysis for 10 journals, 5 departments, or 20 funders. Writing separate scripts for each entity is error-prone and hard to maintain. **Parameterized reports** solve this by defining variables (parameters) that change the report's content without changing its structure.

Quarto supports parameters via the `params` field in the YAML header. Parameters can be strings, numbers, or lists. Inside the document, `params$journal_id` (or `params$institution`) drives data fetching, filtering, and labelling. At render time, parameter values are passed via the command line or a rendering script.

This approach scales naturally: a loop or `purrr::walk()` call renders dozens of reports with different parameters, producing a complete portfolio of bibliometric profiles with consistent methodology and formatting.

Parameterized reports are especially valuable for institutional reporting (annual bibliometric profiles for each department), journal analytics (quarterly reports for a publisher's journal portfolio), and funding assessment (per-grant output summaries).

1.4 Worked example

1.4.1 Defining parameters

A parameterized Quarto document starts with parameters in the YAML header:

```
# Example YAML header (not executable R):
#
# ---
# title: "Journal Bibliometric Profile"
# params:
#   journal_id: "S148561398"
#   journal_name: "Scientometrics"
#   start_year: 2020
#   end_year: 2023
# ---
```

1.4.2 Using parameters in code

```
journal_id <- "S148561398"
journal_name <- "Scientometrics"
start_year <- 2020
end_year <- 2023
```

```
works <- fetch_openalex(
  entity = "works",
  primary_location.source.id = journal_id,
  from_publication_date = paste0(start_year, "-01-01"),
  to_publication_date = paste0(end_year, "-12-31"),
  type = "article",
  options = list(sample = 300, seed = 42)
)

cat(glue("Journal: {journal_name}\n"))
```

```
#> Journal: Scientometrics
```

```
cat(glue("Period: {start_year}--{end_year}\n"))
```

```
#> Period: 2020--2023
```

```
cat(glue("Articles retrieved: {nrow(works)}\n"))
```

```
#> Articles retrieved: 300
```

1.4.3 Automated summary statistics

```
summary_stats <- works |>
  summarise(
    total_articles = n(),
    mean_citations = round(mean(cited_by_count), 1),
    median_citations = median(cited_by_count),
    h_index = compute_h_index(cited_by_count),
    oa_rate = round(mean(oa_status != "closed", na.rm = TRUE) * 100, 1)
  ) |>
```

Metric	Value
total_articles	300
mean_citations	29.6
median_citations	15
h_index	40
oa_rate	51.7

```
  mutate(across(everything(), as.character)) |>
  pivot_longer(everything(), names_to = "Metric", values_to = "Value")

summary_stats |> gt()
```

1.4.4 Parameterized figure

```
works |>
  mutate(year = year(publication_date)) |>
  count(year) |>
  ggplot(aes(x = factor(year), y = n)) +
  geom_col(fill = palette_sci(1)) +
  labs(x = "Year", y = "Articles",
       title = glue("{journal_name} — Annual Output")) +
  theme_sci()
```

1.4.5 Batch rendering

```
# Render for multiple journals:
journals <- tibble(
  journal_id = c("S148561398", "S202381698", "S137773608"),
  journal_name = c("Scientometrics", "PLOS ONE", "Nature")
)

walk2(journals$journal_id, journals$journal_name, function(jid, jname) {
  quarto::quarto_render(
    input = "journal_profile.qmd",
    execute_params = list(
      journal_id = jid,
      journal_name = jname,
      start_year = 2020,
      end_year = 2023
    ),
    output_file = glue("reports/{jname}_profile.html")
  )
})
```

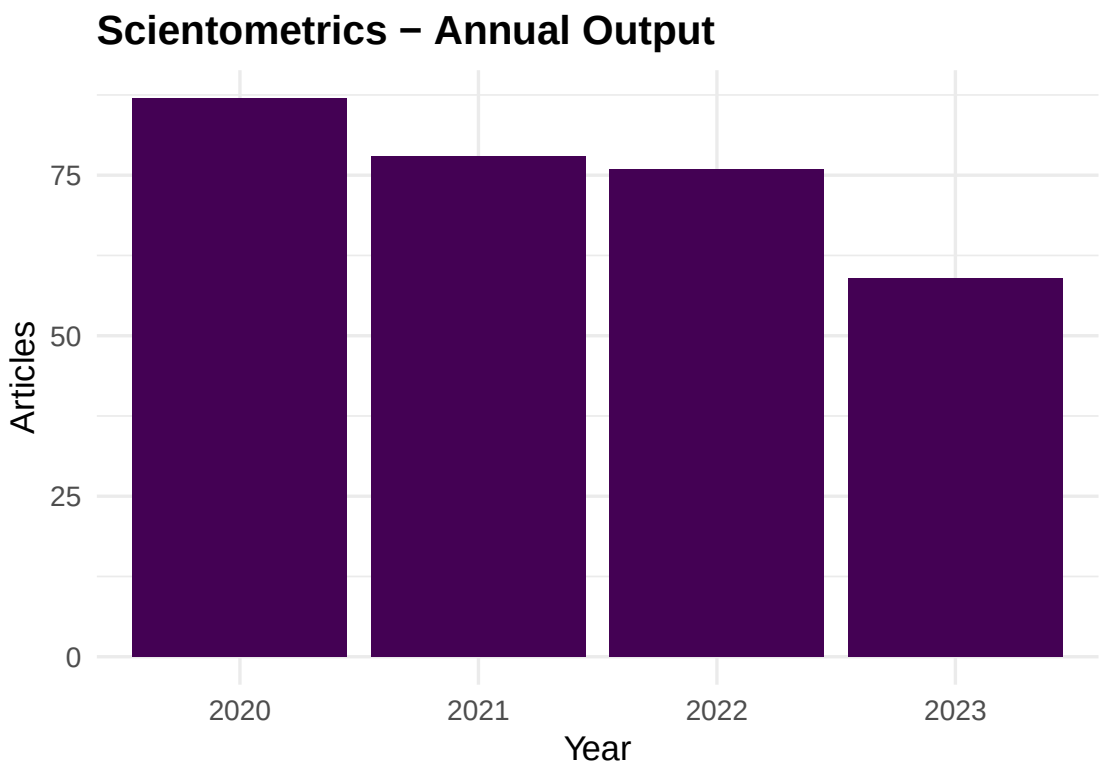


Figure 1: Annual publication count for the selected journal.

1.5 Diagnostics and interpretation

- **Parameter validation**: Check that parameters produce valid API queries. Invalid journal IDs return empty results silently.
- **Sample consistency**: When using `options = list(sample = N)`, the sample changes if the underlying corpus changes. Use `seed` for stability.
- **Output naming**: Automated file naming should avoid special characters. Sanitise journal names before using them in file paths.
- **Error handling**: Batch rendering should catch and log errors for individual reports without stopping the entire batch.

1.6 Limitations and responsible use

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- **Automated ≠ mindless.** Parameterized reports produce consistent output but still require human review. An automated report with a data error will produce the wrong results consistently across all entities.
- **One template fits all?** Different entities may need different analyses. A template designed for a large journal may produce misleading results for a small one (e.g., sparse yearly bins).
- **API costs.** Batch rendering for 100 journals makes 100 API calls. Respect rate limits and cache results.
- **Report fatigue.** Generating too many reports can overwhelm decision-makers. Produce reports for entities that need them, not for every possible parameter value [@hicks2015].

1.8 Common pitfalls

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- **Hardcoding values that should be parameters.** If you find yourself editing a date or ID in the script, it should be a parameter.
- **Not testing with extreme parameters.** A journal with 5 papers per year needs different visualisations than one with 5,000. Test edge cases.
- **Rendering without caching.** Each render re-executes all code. Use Quarto's `cache: true` or file-based caching to avoid redundant API calls.
- **Forgetting output format compatibility.** Interactive widgets work in HTML but not PDF. Ensure your template degrades gracefully.

1.10 Exercises

1. **Institutional template.** Create a parameterized Quarto document for institutional profiles. Parameters: institution ID, name, start year, end year. Include publication count, citation distribution, and top authors.
2. **Batch rendering.** Render the journal profile template for 3 different journals. Compare the resulting reports side by side.
3. **Conditional content.** Add conditional logic: if the sample has fewer than 50 papers, display a warning message instead of the citation distribution figure.

1.11 Solutions

Solutions are provided in **??**.

1.12 Further reading

- @aria2017 — `bibliometrix` reporting features.
- @priem2022 — OpenAlex API for programmatic data retrieval in reports.

1.13 Session info

```
sessionInfo()
```

```
#> R version 4.6.1 (2026-06-24)
#> Platform: x86_64-pc-linux-gnu
#> Running under: Ubuntu 24.04.5 LTS
#>
#> Matrix products: default
#> BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK
 ↪  version 3.12.0
#>
#> locale:
```

```
#>  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8
#>  [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8    LC_MESSAGES=C.UTF-8
#>  [7] LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
#> [10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: UTC
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#>  [1] DT_0.34.0                plotly_4.12.1
#>  [3] uwot_0.2.5               Matrix_1.7-5
#>  [5] word2vec_0.4.1           stm_1.3.8
#>  [7] topicmodels_0.2-17       quanteda.textstats_0.97.2
#>  [9] visNetwork_2.1.4         ggraph_2.2.2
#> [11] tidygraph_1.3.1          igraph_2.3.4
#> [13] quanteda_4.5.1           pdftools_3.9.1
#> [15] arrow_26.0.0             bibliometrix_5.5.0
#> [17] RefManageR_1.4.0         bib2df_1.1.2.0
#> [19] rcrossref_1.2.1          gt_1.4.0
#> [21] tidytext_0.4.3           glue_1.8.1
#> [23] openalexR_3.1.2          lubridate_1.9.5
#> [25] forcats_1.0.1            stringr_1.6.0
#> [27] dplyr_1.2.1              purrr_1.2.2
#> [29] readr_2.2.0              tidyr_1.3.2
#> [31] tibble_3.3.1             ggplot2_4.0.3
#> [33] tidyverse_2.0.0          bookdown_0.48
#> [35] fs_2.1.0                 rmarkdown_2.32
#>
#> loaded via a namespace (and not attached):
#>   [1] bibtex_0.5.3           RColorBrewer_1.1-3     jsonlite_2.0.0
#>   [4] magrittr_2.0.5         modeltools_0.2-25      farver_2.1.2
#>   [7] vctrs_0.7.3            memoise_2.0.1          askpass_1.2.1
#>  [10] base64enc_0.1-6        tinytex_0.61           htmltools_0.5.9
#>  [13] contentanalysis_1.1.1  curl_8.1.0             broom_1.0.13
#>  [16] janeaustenr_1.0.0      cellranger_1.1.0       htmlwidgets_1.6.4
#>  [19] tokenizers_0.3.0       plyr_1.8.9             httr2_1.3.0
#>  [22] cachem_1.1.0           dimensionsR_0.0.3      mime_0.13
#>  [25] lifecycle_1.0.5        pkgconfig_2.0.3        R6_2.6.1
#>  [28] fastmap_1.2.0          shiny_1.14.0           digest_0.6.39
#>  [31] patchwork_1.3.2        shinycssloaders_1.1.0  rprojroot_2.1.1
#>  [34] RSpectra_0.16-2        SnowballC_0.7.1        labeling_0.4.3
#>  [37] urltools_1.7.3.1       timechange_0.4.0       mgcv_1.9-4
#>  [40] httr_1.4.9             polyclip_1.10-7        compiler_4.6.1
#>  [43] here_1.0.2             bit64_4.8.6            withr_3.0.3
#>  [46] backports_1.5.1        S7_0.2.2               viridis_0.6.5
#>  [49] ggforce_0.5.0          MASS_7.3-65            bibliometrixData_0.3.0
#>  [52] tools_4.6.1            otel_0.2.0             stopwords_2.3
#>  [55] zip_3.0.2              rentrez_1.2.4          httpuv_1.6.17
#>  [58] nlme_3.1-169           promises_1.5.0         grid_4.6.1
#>  [61] stringdist_0.9.17      reshape2_1.4.5         generics_0.1.4
#>  [64] gtable_0.3.6           tzdb_0.5.0             rscopus_0.9.0
#>  [67] ca_0.72                data.table_1.18.6.1    hms_1.1.4
#>  [70] xml2_1.6.0             utf8_1.2.6             ggrepel_0.9.8
#>  [73] pillar_1.11.1          nsyllable_1.0.1        vroom_1.7.1
#>  [76] later_1.4.8            splines_4.6.1          tweenr_2.0.3
#>  [79] lattice_0.22-9         FNN_1.1.4.1            bit_4.6.0
#>  [82] tidyselect_1.2.1       tm_0.7-20              miniUI_0.1.2
```

```
#>  [85] knitr_1.52             gridExtra_2.3.1        NLP_0.3-3
#>  [88] stats4_4.6.1           crul_1.6.0             xfun_0.62
#>  [91] graphlayouts_1.2.5     matrixStats_1.5.0      humaniformat_0.6.0
#>  [94] stringi_1.8.9          qpdf_1.4.1             yaml_2.3.12
#>  [97] evaluate_1.0.5         httpcode_0.3.0         cli_3.6.6
#> [100] xtable_1.8-8           Rcpp_1.1.2             readxl_1.6.0
#> [103] triebeard_0.4.1        XML_3.99-0.25          parallel_4.6.1
#> [106] assertthat_0.2.1       pubmedR_1.0.2          slam_0.1-56
#> [109] viridisLite_0.4.3      scales_1.4.0           crayon_1.5.3
#> [112] openxlsx_4.2.9         rlang_1.3.0            fastmatch_1.1-8
```